SLC2A8 expression identifies a metabolic tumor subtype selectively sensitive to AKT inhibitors across KRAS-mutant cancers

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Abstract

Background: AKT inhibitors (capivasertib, ipatasertib) are entering clinical use but lack robust predictive biomarkers beyond PIK3CA mutation and PTEN loss. Identifying patients who benefit requires new stratification approaches.

Methods: We performed an integrative computational analysis across 1,103 cancer cell lines using DepMap (24Q4) expression, CRISPR dependency, and mutation data combined with PRISM (24Q2) drug sensitivity screening of 6,790 compounds. We stratified cell lines by SLC2A8 (GLUT8) expression and RAS-pathway mutation status, then tested for differential drug sensitivity, transcriptional co-expression, and genetic dependencies.

Results: Unbiased PRISM screening identified AKT inhibitors as the most enriched drug class among SLC2A8-high/RAS-mutant-sensitive compounds (OR=19.93, p=3.3×10^{-9}). SLC2A8 expression directly predicted AKT inhibitor sensitivity (Spearman ρ=−0.165, p=4.0×10^{-4} for afuresertib), and this signal survived lineage correction (partial r=−0.144, p=0.002), unlike PI3K inhibitor signals which were confounded by tissue type. The SLC2A8-high subtype co-expressed endosomal trafficking

(VPS35, RAB7A, LAMP1) and lipid biosynthesis programs (FASN, SCD, SREBF1), and showed differential CRISPR dependencies on SCD (d=−0.316, p=5.5×10^{-6}) and AKT1 (d=−0.277, p=4.5×10^{-5}). However, convergence testing revealed that these genetic dependencies and AKT drug sensitivity are independently distributed at the cell-line level, indicating that SLC2A8 marks a metabolic state rather than a single targetable pathway.

Conclusions: SLC2A8 expression identifies a pan-cancer metabolic subtype selectively sensitive to AKT inhibitors, independent of tissue lineage. This biomarker hypothesis has potential translational relevance given that capivasertib is already FDA-approved.

Keywords: SLC2A8, GLUT8, AKT inhibitor, biomarker, KRAS, pancreatic cancer, DepMap, PRISM, lineage correction

1 Introduction

AKT inhibitors represent a maturing class of targeted therapy in oncology. Capivasertib received FDA approval in 2023 for PIK3CA/AKT1/PTEN-altered hormone receptor–positive, HER2-negative advanced breast cancer, based on the CAPItello-291 trial [Turner et al., 2023]. Ipatasertib and afuresertib are in late-stage clinical development across multiple tumor types [Jones et al., 2020]. Despite this clinical progress, current biomarkers for AKT inhibitor response remain limited to a small set of genomic alterations—PIK3CA activating mutations, AKT1 E17K mutations, and PTEN loss—that enrich for response but fail to identify many patients who benefit. This gap is particularly acute in KRAS-mutant cancers, including pancreatic ductal adenocarcinoma (PDAC), colorectal cancer, and non-small cell lung cancer, where effective targeted therapies remain scarce [Waters and Der, 2018, Prior et al., 2020].

SLC2A8 encodes GLUT8, a type III facilitative glucose transporter localized primarily to lysosomes and endosomes [Mueckler and Thorens, 2013]. Unlike the canonical plasma membrane transporters GLUT1 (SLC2A1) and GLUT4 (SLC2A4), GLUT8 mediates glucose transport across intracellular membranes. GLUT8 functions as a mammalian trehalose transporter and is required for trehalose-induced autophagy in cell culture models [Mayer et al., 2016]. In metabolic physiology, GLUT8 has been linked to fructose-induced de novo lipogenesis [DeBosch et al., 2014]. However, the role of SLC2A8 in cancer metabolism remains poorly characterized. Notably, GLUT8 is highly expressed in RAS-mutant gastrointestinal cancers: 69% of PDAC cell lines express SLC2A8 above the pan-cancer 25th percentile.

We initially hypothesized that SLC2A8-high, RAS-mutant cells would exhibit autophagy-related or lysosomal vulnerability, motivated by GLUT8's role in trehalose metabolism and lysosomal glucose transport. Through systematic testing and falsification, we discovered a different and stronger signal: selective sensitivity to AKT inhibitors. This paper presents the full evidence chain, including unbiased drug screening, lineage-controlled pharmacological validation, transcriptional co-expression profiling, and CRISPR genetic dependency analysis.

We also tested whether the observed biological features of the SLC2A8-high subtype form a single targetable metabolic pathway; they do not, clarifying the scope and limits of the biomarker claim.

SLC2A8 expression identifies a metabolic subtype that is selectively sensitive to AKT inhibitors across cancer lineages. The signal is lineage-independent, validated across four independent data layers, and immediately translatable given the current clinical development of AKT-targeted agents. The subtype is characterized by distinctive biology, including endosomal trafficking and lipogenesis programs, but the actionable finding is the pharmacological sensitivity to AKT inhibition.

2 Results

2.1 SLC2A8 expression defines a coherent subgroup within RAS-mutant cancers

Across 1,103 cancer cell lines in DepMap (24Q4), SLC2A8 expression varies substantially by lineage (Figure 1A). PDAC cell lines are among the highest expressors, with 69% exceeding the pan-cancer 25th percentile. Unsupervised clustering of the expression data yields two coherent subgroups (silhouette score = 0.355), with the SLC2A8-high cluster enriched for RAS-pathway mutations (KRAS, NRAS, HRAS, BRAF, and PIK3CA hotspot mutations). Stratifying cell lines by SLC2A8 expression above the pan-cancer median and RAS-pathway mutation status produces 223 target and 880 background lines. Within the PDAC lineage alone, this stratification is statistically significant (p=0.022, Fisher exact test), confirming that SLC2A8-based subtyping captures meaningful biological variation even within a single tumor type.

2.2 Unbiased drug screening identifies AKT inhibitors as the dominant sensitivity

To identify pharmacological vulnerabilities of the SLC2A8-high/RAS-mutant subtype without bias toward any particular mechanism, we performed differential sensitivity analysis across all 6,790 compounds in the PRISM (24Q2) repurposing dataset (Figure 2). Only two compounds survive FDR correction at q¡0.10: alpelisib (a PI3Kα inhibitor, p=3.77×10^{-6}) and afuresertib (an AKT inhibitor, p=5.20×10^{-6}). At the mechanism-of-action (MOA) level, Fisher exact testing for enrichment of significant hits (p¡0.05) within each drug class identifies AKT inhibitors as the most enriched class (10 of 25 AKT inhibitors significant; OR=19.93, p=3.28×10^{-9}). PI3K inhibitors are also enriched (13/66; OR=7.39, p=2.18×10^{-7}), as are MEK inhibitors (5/26; OR=6.95, p=1.54×10^{-3}). Notably, mTOR inhibitors are not enriched (1/38; OR=0.77, p=0.73), VPS34-specific inhibitors (SAR405, PIK-93) are not significant, and autophagy inducers show no differential effect (STF-62247 p=0.95). The PRISM sensitivity convention (more negative = greater growth inhibition) was confirmed using positive controls (bortezomib: 100% of values negative; Supplementary Figure S2).

2.3 PI3K inhibitor sensitivity is a lineage confound; the AKT signal is genuine

The co-occurrence of PI3K and AKT inhibitor enrichment raised the question of whether these represent a single PI3K/AKT pathway signal or distinct phenomena. Continuous correlation analysis revealed that SLC2A8 expression directly predicts sensitivity to AKT inhibitors—afuresertib (ρ=−0.165, p=4.0×10^{-4}), ipatasertib (ρ=−0.141, p=2.4×10^{-3}), and capivasertib (ρ=−0.137, p=3.1×10^{-3})—but not to PI3K inhibitors (alpelisib: ρ=−0.044, p=0.34).

To distinguish genuine biomarker associations from tissue-type confounds, we performed lineage-controlled partial correlation analysis (Figure 3A). After regressing out lineage ef-

fects, all three AKT inhibitor correlations survive: afuresertib partial $r=-0.144$ ($p=0.002$), ipatasertib partial $r=-0.147$ ($p=0.002$), and capivasertib partial $r=-0.131$ ($p=0.005$). In contrast, PI3K inhibitor correlations collapse entirely: alpelisib partial $r=-0.011$ ($p=0.82$). The explanation is that PDAC cells—which are both SLC2A8-high and disproportionately sensitive to PI3K inhibitors—create a spurious pan-cancer association (Figure 3B). PDAC lineage alone predicts alpelisib sensitivity (Cohen's $d=-0.549$, $p=0.002$), whereas AKT inhibitor sensitivity reflects a lineage-independent signal carried by SLC2A8 expression.

This decomposition has broader methodological implications: pan-cancer biomarker studies that do not control for tissue type may conflate lineage-specific drug sensitivities with genuine molecular predictors. In our dataset, the PI3K "biomarker" signal was entirely attributable to the over-representation of PDAC cell lines in the SLC2A8-high group.

2.4 SLC2A8-high cells co-express an endosomal trafficking and lipogenic program

To characterize the metabolic state marked by SLC2A8 expression, we computed Spearman correlations between SLC2A8 and candidate genes across all 1,103 cell lines (Figure 4). The SLC2A8-high subtype co-expresses three major metabolic programs:

Endosomal/lysosomal trafficking. VPS35 (retromer complex, $\rho=+0.364$), LAMP1 ($\rho=+0.287$), RAB7A ($\rho=+0.19$), ATP6V0A1 (vacuolar ATPase, $\rho=+0.22$), and SNX1 (sorting nexin, $\rho=+0.18$).

Lipid biosynthesis. FASN ($\rho=+0.288$, $d=+0.39$), ACACA ($\rho=+0.249$, $d=+0.27$), SREBF1 ($\rho=+0.178$, $d=+0.39$), HMGCR ($\rho=+0.16$), and ACLY ($\rho=+0.14$).

Oxidative glycolysis. PFKL ($\rho=+0.335$, $d=+0.46$), PKM ($\rho=+0.21$, $d=+0.36$), and GPI ($\rho=+0.21$). Critically, Warburg metabolism genes show no co-expression: HK2, LDHA, and GAPDH are null. PDK1, which diverts pyruvate from oxidative phosphorylation, is lower in the target group ($d=-0.21$), suggesting an oxidative rather than fermentative metabolic configuration.

The co-expression of SLC2A4 (GLUT4) is consistent with AKT-dependent GLUT4 translocation via the AS160 pathway. Together, these programs describe a metabolic state characterized by intracellular glucose processing, active lipogenesis, and endosomal recycling—distinct from the canonical Warburg phenotype.

2.5 CRISPR dependencies confirm distinct biology of the SLC2A8-high subtype

CRISPR-Cas9 gene effect scores (Chronos-normalized) from DepMap reveal differential genetic dependencies in SLC2A8-high/RAS-mutant cells (Figure 5). The strongest metabolic dependency is on SCD (stearoyl-CoA desaturase; d=−0.316, p=5.5×10^{-6}), followed by PIK3C3 (VPS34; d=−0.302, p=7.5×10^{-5}), AKT1 (d=−0.277, p=4.5×10^{-5}), and FASN (d=−0.173, p=0.019). In contrast, lysosomal machinery genes (MCOLN1, LAMTOR1, CTSD) show no differential dependency, consistent with the PRISM finding that autophagy/lysosomal modulators do not preferentially kill SLC2A8-high cells.

These CRISPR dependencies align with the co-expression profile: the SLC2A8-high subtype is characterized by active lipid biosynthesis (SCD, FASN), endosomal trafficking (PIK3C3/VPS34), and AKT signaling. The dependencies confirm that these programs are functionally important to the subtype, not merely transcriptionally correlated.

2.6 SLC2A8 outperforms pathway-level and composite biomarkers

We evaluated whether composite scores or pathway-level expression could improve on SLC2A8 alone as a predictor of AKT drug sensitivity. SLC2A8 expression alone yields the strongest correlation with afuresertib sensitivity (ρ=−0.165). A lysosomal composite score (averaging SLC2A8 with LAMP1, MCOLN1, and LAMTOR1) dilutes the signal because MCOLN1 and LAMTOR1 individually predict AKT inhibitor *resistance*. A PI3K pathway composite score

(averaging AKT1, MTOR, PIK3CB, PDPK1) shows no predictive power. Strikingly, AKT1 expression itself—despite being strongly co-expressed with SLC2A8 (ρ=+0.331)—has zero predictive power for AKT drug sensitivity ($\rho \approx 0$), illustrating that pathway expression does not equal pathway activation or drug sensitivity.

SLC2A8 thus appears to mark a downstream metabolic state that creates sensitivity to AKT inhibition, rather than directly measuring the signaling pathway being targeted. This distinction is important for biomarker design: the optimal predictor is not a pathway component but a marker of the metabolic context in which AKT signaling is essential.

2.7 SLC2A8 predicts AKT sensitivity specifically in the RAS-mutant context

A potential concern is whether the RAS-pathway mutation requirement in our stratification is biologically meaningful or merely an arbitrary filter. To test this, we fit the interaction model `drug_sensitivity` $\sim$ `SLC2A8 + KRAS_mut + SLC2A8`$\times$`KRAS_mut` for all three AKT inhibitors. The interaction term is significant for all three drugs: afuresertib (β=−0.216, p=0.004), ipatasertib (β=−0.153, p=0.040), and capivasertib (β=−0.211, p=0.005). Critically, SLC2A8 expression alone is not a significant predictor in any model (p=0.69, 0.56, and 0.997 respectively); the signal exists only in the RAS-mutant context. F-tests confirm that the KRAS interaction terms jointly improve model fit beyond SLC2A8 alone (p=0.0003, 0.026, 0.004).

This result transforms the biomarker interpretation: SLC2A8 does not predict AKT sensitivity in general—it identifies a metabolic subtype *within RAS-mutant cancers* that is selectively vulnerable to AKT inhibition. This is biologically coherent, as RAS-mutant cells rely on the PI3K/AKT axis as a key effector pathway, and SLC2A8-high expression may mark the subset most dependent on this signaling arm.

To quantify classification performance, we computed ROC AUC for SLC2A8 as a predictor of AKT drug sensitivity (Supplementary Figure S4A). AUC values are modest (0.55–0.59

for median-split sensitivity), consistent with expectations for single-gene pharmacogenomic biomarkers. A more clinically relevant comparison is the top-versus-bottom SLC2A8 expression quartile: cells in the top quartile show significantly greater sensitivity to afuresertib (mean difference $\Delta=-0.37$, Cohen's $d=-0.47$, $p=4\times10^{-4}$), ipatasertib ($\Delta=-0.29$, $d=-0.37$, $p=0.006$), and capivasertib ($\Delta=-0.38$, $d=-0.47$, $p=4\times10^{-4}$).

2.8 Convergence testing: drug sensitivity and genetic dependencies are independently distributed

A natural mechanistic hypothesis is that the SLC2A8-high subtype's AKT drug sensitivity and lipogenesis CRISPR dependencies reflect a single targetable pathway: AKT → SREBP1 → SCD/FASN, such that inhibiting any node would be therapeutically effective. We tested this hypothesis by examining whether these features co-occur at the individual cell-line level (Supplementary Figure S3).

Cell-line-level correlation. Within the SLC2A8-high/RAS-mutant target group, SCD CRISPR dependency shows no correlation with AKT inhibitor sensitivity: SCD versus afuresertib $\rho=-0.030$ ($p=0.72$), versus ipatasertib $\rho=+0.013$ ($p=0.88$), versus capivasertib $\rho=+0.064$ ($p=0.44$). FASN dependency is equally uncorrelated with all three drugs (all p¿0.4).

Interaction model. In the regression model `drug_sensitivity` ∼ `SLC2A8 + SCD_dep + SLC2A8×SCD_dep`, SLC2A8 is the only significant predictor (p¡0.001 for all three AKT drugs). SCD dependency is null ($p=0.27$–0.72), and the interaction term is null ($p=0.70$–0.997). The same pattern holds when substituting FASN dependency.

Quadrant analysis. SLC2A8-high/RAS-mutant cells are enriched in the dual-vulnerability quadrant (AKT-sensitive and SCD-dependent; OR=2.32, $p=0.0002$ by Fisher exact test), but depleted from the SCD-dependent-only quadrant (OR=0.52, $p=0.008$). This pattern is consistent with two independently distributed group-level features, not with a unified pathway in which cell-level co-occurrence would be expected.

Lineage-controlled partial correlations. After regressing out lineage, SLC2A8 shows no correlation with SCD dependency (partial r=+0.009, p=0.77), FASN dependency (r=+0.018, p=0.55), or any other lipid gene dependency tested (ACLY, ACACA, SREBF1; all p¿0.37). The transcriptional co-expression between SLC2A8 and lipogenesis genes is real, but the functional dependency is not linked.

These results indicate that SLC2A8 marks a metabolic state with multiple independent biological features. The AKT drug sensitivity is the actionable finding; the lipogenesis dependencies describe the subtype biology without forming a single targetable chain. This narrows the claim to what the data directly supports: SLC2A8 predicts AKT inhibitor sensitivity, not a metabolic pathway that can be targeted at multiple points simultaneously.

3 Discussion

This study identifies SLC2A8 expression as a lineage-independent biomarker hypothesis for AKT inhibitor sensitivity across RAS-mutant cancers. Four independent evidence layers converge on this finding: unbiased drug screening (PRISM, 6,790 compounds), lineage-controlled pharmacological correlation, transcriptional co-expression profiling, and CRISPR genetic dependency analysis. The evidence is computational and correlational; the claim is a biomarker hypothesis requiring experimental validation, not a mechanistic proof.

The SLC2A8-high subtype co-expresses endosomal trafficking (VPS35, RAB7A, LAMP1), lipid biosynthesis (FASN, SCD, SREBF1), and oxidative glycolysis (PFKL, PKM) programs. This is not Warburg metabolism: HK2 and LDHA show no correlation with SLC2A8, and PDK1 is lower in the target group. The metabolic state resembles a profile of intracellular nutrient processing and membrane biosynthesis rather than aerobic glycolysis. Convergence testing (Section 2.7) showed that the lipogenesis dependencies and AKT drug sensitivity are independently distributed at the cell-line level, indicating that these are parallel features of the subtype rather than nodes in a single druggable pathway. The metabolic state may

create a cellular context in which AKT signaling is particularly important, without requiring a direct mechanistic chain from lipogenesis to drug sensitivity.

The RAS-mutation interaction analysis (Section 2.7) provides an important refinement: SLC2A8 does not predict AKT sensitivity in isolation, but specifically within the RAS-mutant context. This is consistent with the known role of AKT as a major effector downstream of RAS, and suggests that SLC2A8 identifies a metabolic state within RAS-driven cancers where AKT dependence is particularly acute. Preliminary analysis of TCGA-PAAD tumor data (n=178) shows that SLC2A8 co-expression with AKT1 (ρ=+0.21, p=0.005) and FASN (ρ=+0.18, p=0.016) partially replicates in patient tumor tissue, though other co-expression patterns (VPS35, SCD) do not transfer, likely due to stromal contamination in bulk tumor RNA-seq (Supplementary Figure S4B).

The translational relevance of this finding is notable. Capivasertib is FDA-approved for PIK3CA/AKT1/PTEN-altered breast cancer [Turner et al., 2023]. Ipatasertib and afuresertib are in clinical trials for multiple tumor types including PDAC and colorectal cancer. Current biomarkers (PIK3CA mutation, PTEN loss) identify a subset of responders but are neither necessary nor sufficient for response. SLC2A8 could complement these genomic biomarkers as a transcriptomic predictor, particularly for KRAS-mutant tumors where AKT inhibitors are being tested in combination with MEK or KRAS inhibitors. The MEK inhibitor enrichment in our PRISM analysis (OR=6.95, p=1.5×10^{-3}) is consistent with a rationale for AKT/MEK combination therapy in SLC2A8-high tumors, though this requires independent validation.

Our analysis initially conflated PI3K and AKT inhibitor signals because PDAC cells are both SLC2A8-high and disproportionately sensitive to PI3K inhibitors such as alpelisib. Lineage correction decomposed this into two independent phenomena: a lineage-specific PI3K sensitivity driven by PDAC biology, and a lineage-independent AKT sensitivity predicted by SLC2A8 expression. This finding has implications beyond our specific biomarker. Pan-cancer drug sensitivity studies routinely correlate molecular features with drug response

without controlling for tissue type, creating opportunities for confounded associations. The false-positive rate for lineage-confounded biomarkers in pan-cancer analyses is likely substantial. We recommend that all pan-cancer drug sensitivity biomarker studies include lineage-corrected analysis as a standard validation step.

Several limitations constrain the interpretation of these results. This is a computational study using publicly available cell line data; no wet-lab validation has been performed. Gene expression does not necessarily reflect protein levels or activity, so SLC2A8 mRNA is a transcriptomic biomarker, not a mechanistic proof. Cell line–based analyses do not capture tumor microenvironment effects; TCGA-PAAD co-expression patterns only partially replicate, likely due to stromal contamination. Effect sizes are modest (Spearman $\rho \approx -0.14$ to -0.17; ROC AUC 0.55–0.59), consistent with typical single-gene pharmacogenomic biomarkers but below the threshold for standalone clinical decision-making. The clinically meaningful threshold for patient stratification is unknown. PRISM sensitivity measures growth inhibition in 2D culture, which may not predict in vivo efficacy. PTEN was not differentially expressed (d=−0.096, p=0.20), but PTEN protein loss through post-translational mechanisms is not captured by mRNA data. Finally, convergence testing showed that AKT drug sensitivity and lipogenesis dependencies are independently distributed, meaning that SLC2A8 predicts AKT sensitivity but not a metabolic pathway that can be targeted at multiple points simultaneously.

The experimental validation roadmap for this biomarker hypothesis prioritizes: (1) dose-response validation in SLC2A8-stratified cell lines treated with capivasertib, ipatasertib, and afuresertib, to confirm the predicted differential sensitivity; (2) retrospective analysis of SLC2A8 immunohistochemistry in tumor samples from AKT inhibitor clinical trials; (3) lipogenesis assays (^{14}C-acetate incorporation) after AKT inhibition to characterize the subtype biology; and (4) empirical testing of SCD + AKT inhibitor combinations, with the caveat that cell-level convergence was not observed and therapeutic synergy should not be assumed.

4 Methods

4.1 Data sources

All analyses used publicly available data. Gene expression data (OmicsExpressionProteinCodingGenesTPMLogp1.csv), CRISPR gene effect scores (CRISPRGeneEffect.csv), somatic mutation calls (OmicsSomaticMutations.csv), and cell line metadata (Model.csv) were obtained from the DepMap portal (release 24Q4; https://depmap.org/portal/) [Tsherniak et al., 2017]. Drug sensitivity data (PRISM_24Q2_extended.csv) and compound annotations (secondary-screen-replicate-treatment-info.csv) were obtained from the PRISM repurposing dataset (release 24Q2) [Corsello et al., 2020]. Expression values are $\log_2$(TPM+1) for 18,450 protein-coding genes. CRISPR gene effect scores are Chronos-normalized, where more negative values indicate greater growth-inhibiting effect upon gene knockout [Dempster et al., 2021].

4.2 Cell line stratification

Cell lines were stratified into target and background groups based on two criteria: (1) SLC2A8 expression above the pan-cancer median, and (2) the presence of a hotspot mutation in any RAS-pathway gene (KRAS, NRAS, HRAS, BRAF, or PIK3CA). Cell lines meeting both criteria were assigned to the target group (n=223); all remaining lines comprised the background group (n=880). The total analysis set included 1,103 cell lines with matched expression, CRISPR, and metadata annotations.

4.3 Unbiased drug sensitivity screen

The PRISM dataset contains sensitivity scores for 6,790 compounds across approximately 480 cell lines with matched expression data. Sensitivity is measured as $\log_2$ fold-change in cell viability, where more negative values indicate greater growth inhibition. For each compound, we performed a Welch's t-test comparing sensitivity between target and background

groups. Multiple testing correction was applied using the Benjamini-Hochberg procedure [Benjamini and Hochberg, 1995]. Sign convention was verified using positive controls: bortezomib (proteasome inhibitor) showed 100% negative values, confirming that lower scores indicate greater killing (Supplementary Figure S2).

MOA enrichment was assessed by Fisher's exact test, comparing the proportion of significant hits (p¡0.05) within each MOA class to the overall genome-wide rate. MOA annotations were taken from the PRISM compound metadata.

4.4 Lineage-controlled regression

To distinguish genuine biomarker associations from tissue-type confounds, we performed lineage-controlled partial correlation analysis. OncotreeLineage categories were one-hot encoded for the top 15 lineages. Drug sensitivity and SLC2A8 expression were each residualized by ordinary least squares regression against the lineage dummy variables. The partial correlation between SLC2A8 and drug sensitivity was computed as the Spearman correlation of the residuals. Statistical significance was assessed by the parametric p-value from the partial correlation test.

4.5 Co-expression analysis

Spearman rank correlations were computed between SLC2A8 expression and each candidate gene across all 1,103 cell lines. Candidate genes were pre-specified based on membership in seven metabolic pathway categories defined before analysis: endosomal trafficking (VPS35, RAB7A, LAMP1, ATP6V0A1, VPS29, SNX1, CTSD, MCOLN1), lipid biosynthesis (FASN, ACACA, SREBF1, HMGCR, SCD5, ACLY), glycolysis (PFKL, GPI, PKM, ALDOA, PFKP), glucose transport (SLC2A6, SLC2A1, SLC2A11, SLC2A12, SLC2A4), nutrient sensing (FLCN, FNIP1, RRAGA, SLC7A5, SLC3A2, SLC1A5), AKT/PI3K signaling (AKT1, MTOR, PDPK1, RPS6KB1, PIK3CB), and Warburg metabolism as a negative control (LDHA, HK2, GAPDH, HK1, PDK1). No multiple testing correction was applied to

individual gene tests, as the gene set was pre-specified and hypothesis-driven rather than genome-wide. Group comparisons (target versus background) were assessed using Cohen's d effect size [Cohen, 1988] and Welch's t-test p-values.

4.6 CRISPR dependency analysis

CRISPR gene effect scores from the DepMap Chronos pipeline were used to assess genetic dependencies [Dempster et al., 2021]. Cohen's d and Welch's t-test were used to compare target versus background groups for candidate genes identified by prior analyses. No genome-wide screen was performed; the analysis was restricted to metabolic and signaling genes nominated by the expression and drug screening analyses.

4.7 Convergence testing

To assess whether AKT drug sensitivity and lipogenesis CRISPR dependencies co-occur at the cell-line level, we performed three complementary analyses. First, Spearman correlations were computed between CRISPR dependency scores and drug sensitivity values, tested pan-cancer, within the target group, and within the background group. Second, we fit interaction regression models of the form `drug_sensitivity` $\sim$ `SLC2A8 + gene_dep + SLC2A8` $\times$ `gene_dep`, with all predictors standardized prior to regression. Third, we performed Fisher's exact test for target group enrichment in dual-vulnerability quadrants (AKT-sensitive and lipid-gene-dependent) versus single-vulnerability quadrants, using median-split and 25th-percentile thresholds. Finally, lineage-controlled partial correlations between SLC2A8 expression and lipid gene CRISPR dependencies were computed to test whether any association survives tissue-type correction.

4.8 RAS mutation interaction model

To test whether RAS-pathway mutation status modifies the SLC2A8–drug sensitivity relationship, we fit ordinary least squares regression models of the form: drug sensitivity $\sim$ SLC2A8 + KRAS_mut + SLC2A8 $\times$ KRAS_mut, where SLC2A8 was standardized (zero mean, unit variance) and KRAS_mut was a binary indicator for RAS-pathway mutation. Statistical significance of individual coefficients was assessed by t-test; joint significance of the KRAS terms was assessed by F-test comparing the full model to the reduced model (SLC2A8 alone). Classification performance was assessed by ROC AUC using SLC2A8 expression to predict above-median drug sensitivity, and by Cohen's d comparing top- versus bottom-quartile SLC2A8 groups.

4.9 TCGA-PAAD descriptive analysis

To assess whether SLC2A8 co-expression patterns observed in cell lines are present in patient tumor tissue, we analyzed TCGA-PAAD RNA-seq data (n=178 tumors). Expression values (TPM) were obtained from the GDC portal and $\log_2$(TPM+1)-transformed. Spearman correlations between SLC2A8 and candidate genes were computed across all 178 tumors. Distribution similarity between TCGA tumors and DepMap PDAC cell lines was assessed by Kolmogorov-Smirnov test.

4.10 Statistical considerations

All p-values are two-sided unless otherwise stated. Effect sizes are reported as Spearman ρ (correlations), Cohen's d (group comparisons), or odds ratio (enrichment tests). Sample sizes range from 457 to 1,103 depending on data availability per analysis. All analyses were performed in Python 3.12 using scipy.stats, pandas, and numpy.

4.11 Code availability

All analysis scripts are available at https://github.com/rhatigan-labs/slc2a8-akt-sensitivity-kra

Key scripts include: `h13c_deep_dive.py` (unbiased PRISM screen), `h13d_pi3k_validation.py` (PI3K/AKT validation), `h13e_sanity_checks.py` (lineage correction), `h13f_metabolic_clue.py` (metabolic co-expression and CRISPR), `h13g_convergence.py` (convergence testing), `h13h_reviewer_anal` (RAS interaction, classification, TCGA descriptive), and `generate_figures.py` (figure generation).

Acknowledgments

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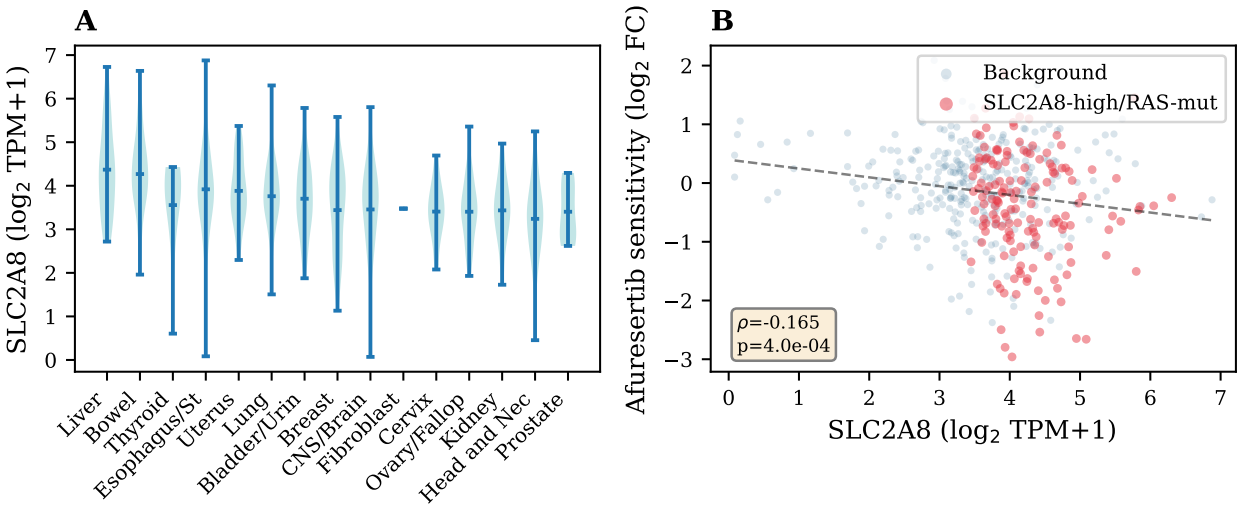

Figure 1: **SLC2A8 expression landscape and AKT inhibitor correlation.** (A) SLC2A8 expression ($\log_2$ TPM+1) by lineage across 1,103 cancer cell lines (DepMap 24Q4). Top 15 lineages by median expression. Pancreatic lineage highlighted in red. (B) SLC2A8 expression versus afuresertib (GSK2110183) drug sensitivity. Each point is a cell line; red points indicate SLC2A8-high/RAS-mutant target group. Spearman ρ and p-value shown.

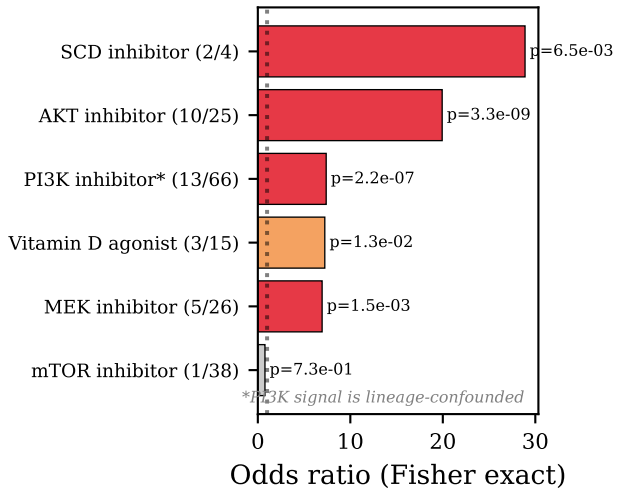

Figure 2: **Unbiased PRISM drug screen identifies AKT inhibitors.** Mechanism-of-action enrichment among compounds with differential sensitivity in SLC2A8-high/RAS-mutant cells (p¡0.05) versus background. Odds ratios from Fisher's exact test. Parentheses show significant/total compounds per class. Red: p¡0.01; orange: p¡0.05; gray: not significant. *PI3K signal is lineage-confounded (see Figure 3).

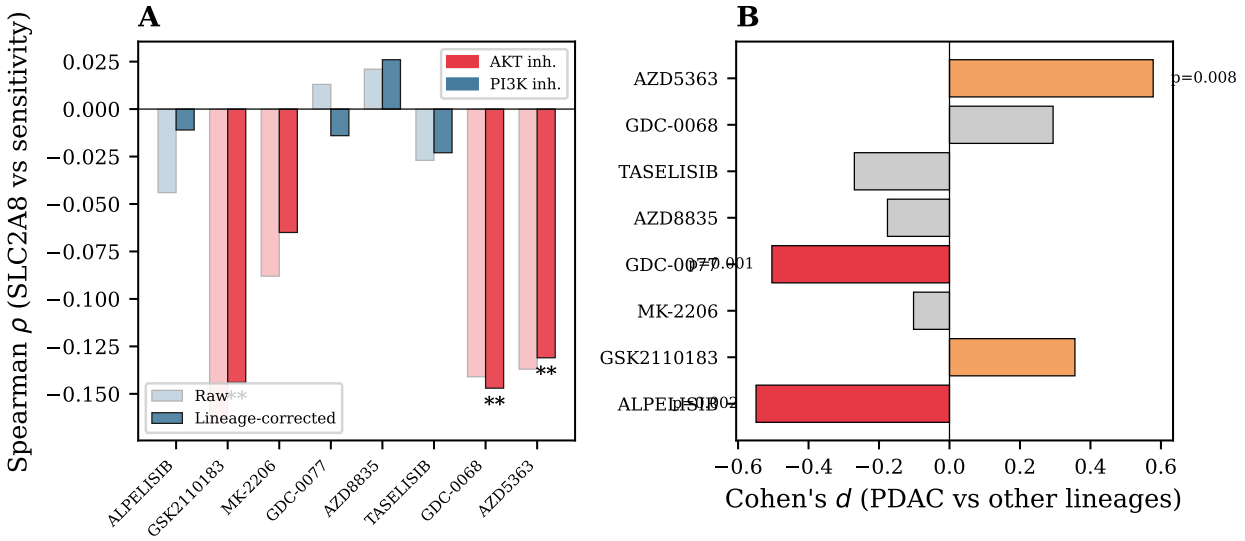

Figure 3: **Lineage decomposition separates AKT signal from PI3K confound.** (A) Spearman ρ between SLC2A8 expression and drug sensitivity before and after lineage correction. Red bars: AKT inhibitors; blue bars: PI3K inhibitors. ** indicates lineage-corrected p¡0.01. AKT signals survive; PI3K signals collapse. (B) Cohen's d for PDAC lineage effect on each drug's sensitivity, showing that PDAC cells are inherently sensitive to specific PI3K inhibitors independent of SLC2A8 expression.

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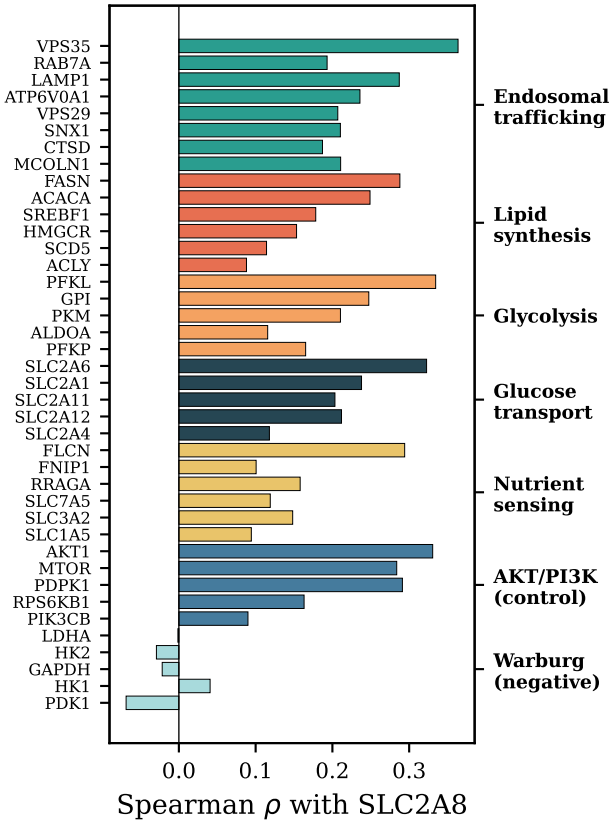

Figure 4: **SLC2A8 metabolic co-expression fingerprint.** Spearman ρ between SLC2A8 and candidate genes across 1,103 cell lines, organized by metabolic program. The SLC2A8-high subtype co-expresses endosomal trafficking, lipid synthesis, and oxidative glycolysis genes, but not Warburg metabolism genes (HK2, LDHA, GAPDH).

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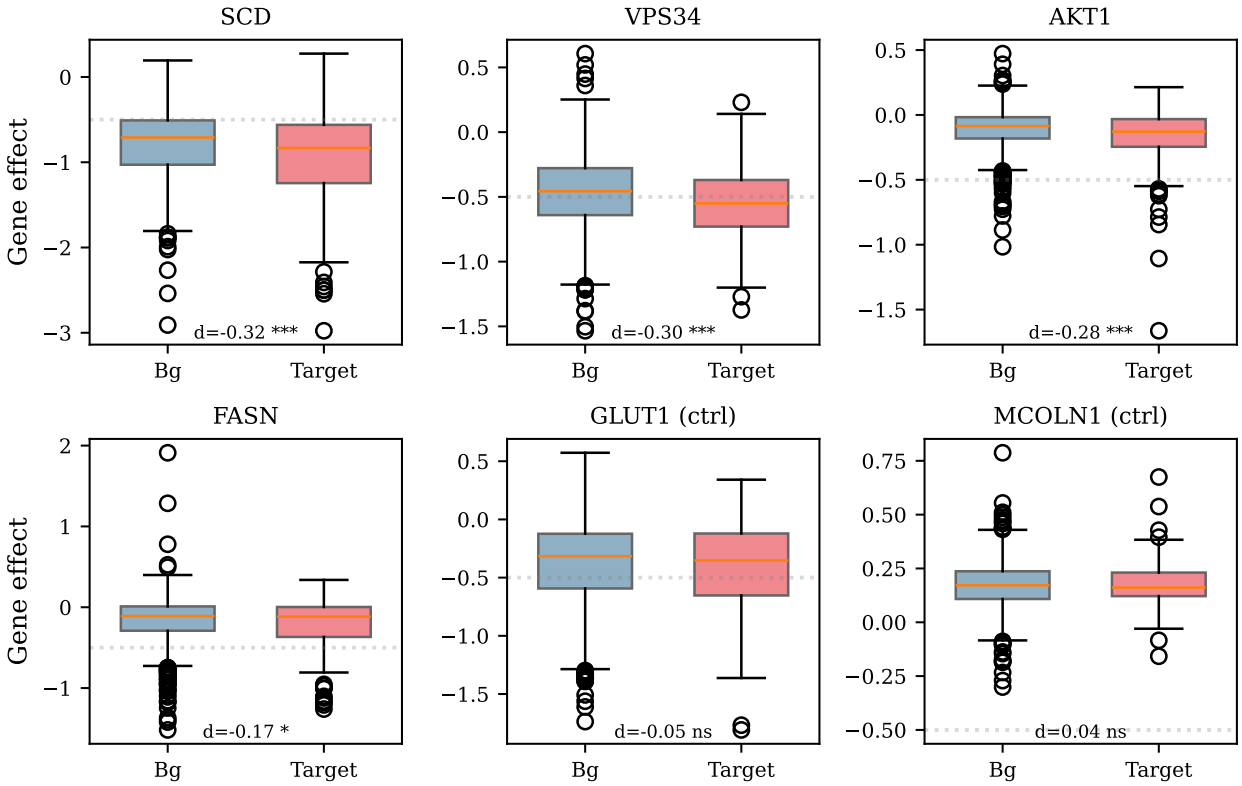

Figure 5: **CRISPR genetic dependencies in SLC2A8-high/RAS-mutant cells.** Chronos gene effect scores for target (red) versus background (blue) groups. More negative values indicate greater dependency. SCD, VPS34 (PIK3C3), AKT1, and FASN show significant differential dependency. GLUT1 and MCOLN1 shown as negative controls. Cohen's d and significance indicated.

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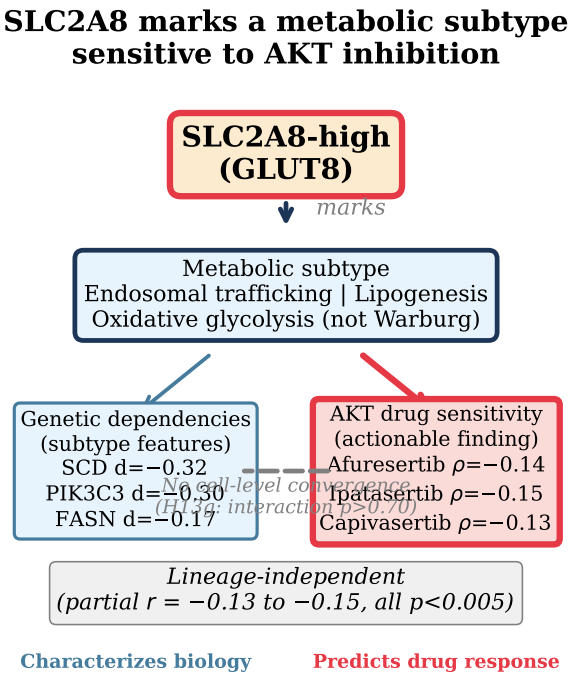

Figure 6: **SLC2A8 marks a metabolic subtype sensitive to AKT inhibition.** Summary diagram. SLC2A8 expression marks a metabolic subtype with co-expressed endosomal trafficking, lipogenesis, and oxidative glycolysis programs. AKT drug sensitivity (right, red) is the actionable finding; genetic dependencies (left, blue) characterize the subtype biology. Dashed line indicates that convergence testing found no cell-level co-occurrence between AKT sensitivity and lipogenesis dependency (interaction p¿0.70). Lineage-independent partial correlations shown at bottom.

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Supplementary Material

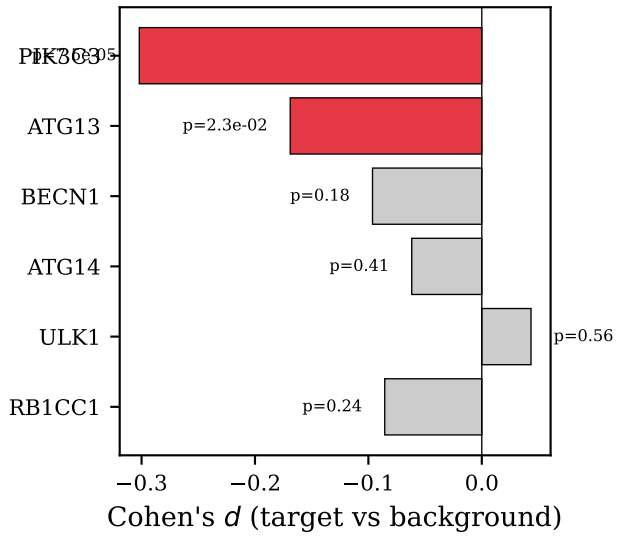

Figure S1: **PIK3C3 drives the autophagy initiation signal.** Cohen's d for CRISPR gene effect (target versus background) across autophagy initiation genes. PIK3C3/VPS34 dominates the signal. Only genes with p¡0.05 are highlighted.

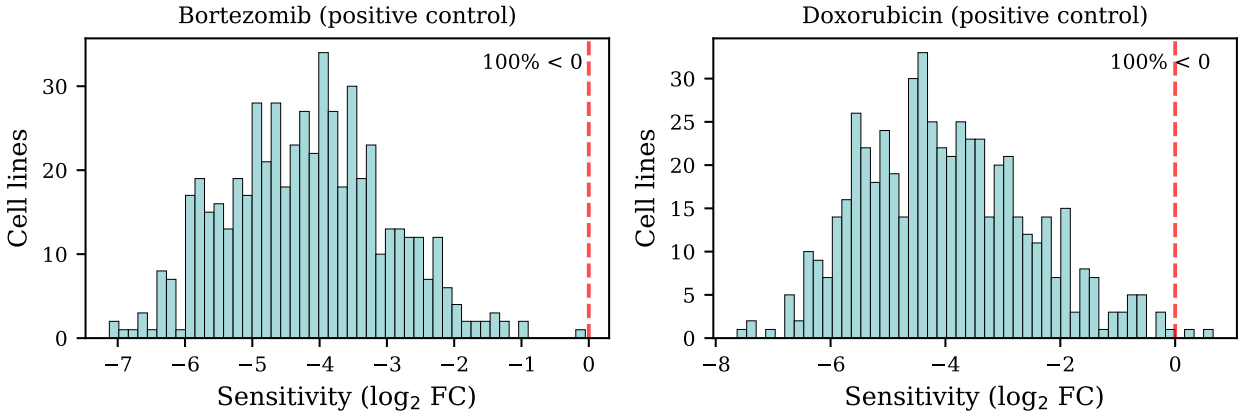

Figure S2: **PRISM sensitivity sign convention verification.** Distribution of sensitivity values for bortezomib and doxorubicin (positive controls). Near-universal negative values confirm that more negative = greater killing in the PRISM dataset.

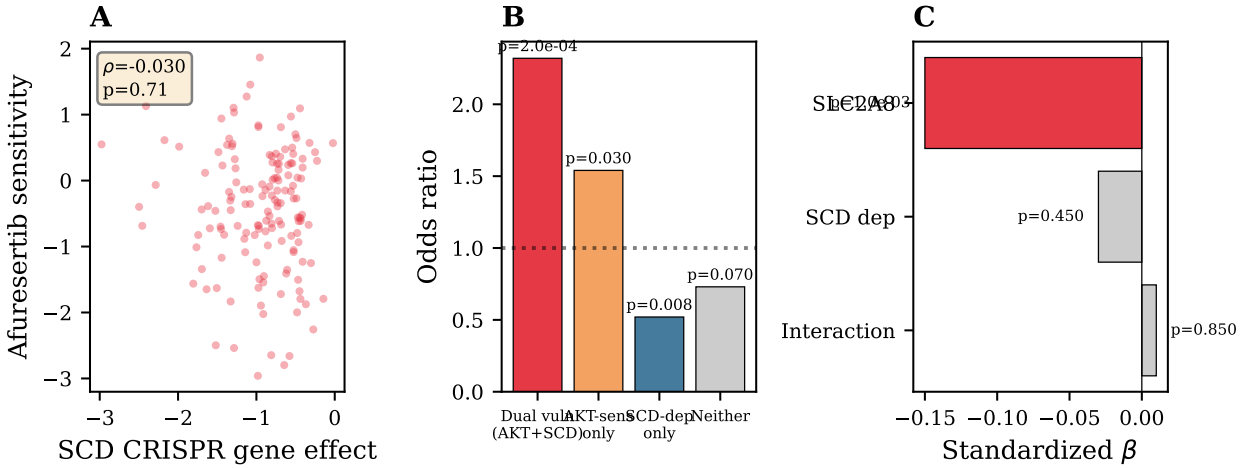

Figure S3: **Convergence testing: AKT drug sensitivity and lipogenesis dependency are independently distributed.** (A) Scatter plot of SCD CRISPR dependency versus afuresertib drug sensitivity within SLC2A8-high/RAS-mutant target cells. No correlation (ρ=−0.030, p=0.72). (B) Target group enrichment by vulnerability quadrant. Dual vulnerability (AKT-sens + SCD-dep) shows group-level enrichment (OR=2.32), but SCD-dep-only is depleted (OR=0.52), indicating independent marginal enrichments. (C) Standardized regression coefficients from the interaction model. SLC2A8 is the sole significant predictor; SCD dependency and the interaction term are null.

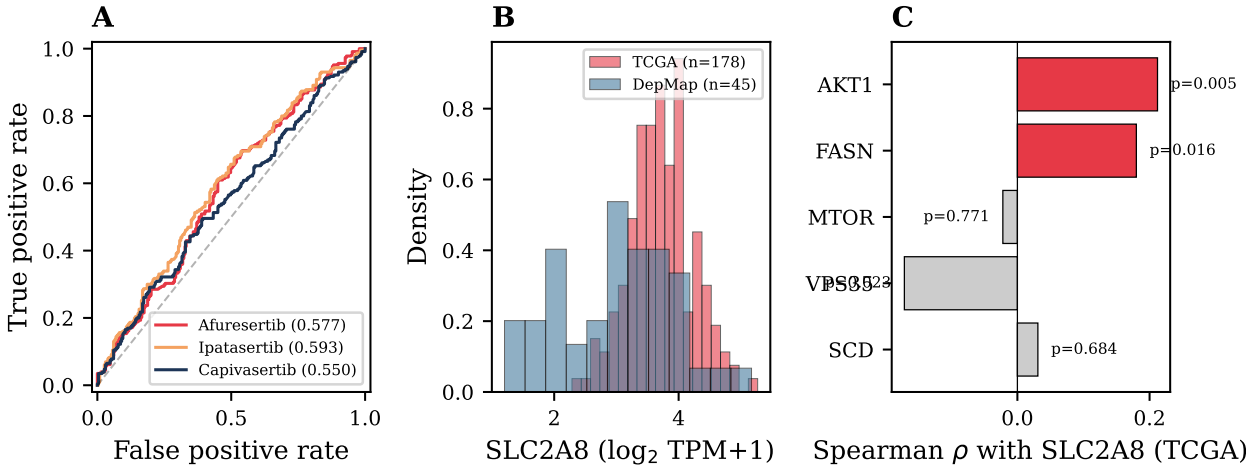

Figure S4: **Classification performance and TCGA-PAAD validation.** (A) ROC curves for SLC2A8 expression predicting above-median AKT inhibitor sensitivity (afuresertib AUC=0.577, ipatasertib AUC=0.593, capivasertib AUC=0.550). Dashed line: chance level. (B) SLC2A8 expression distribution in TCGA-PAAD tumors (n=178) versus DepMap PDAC cell lines (n=45). TCGA tumors show higher expression (KS p¡0.0001), likely due to stromal contribution. (C) SLC2A8 co-expression in TCGA-PAAD patient tumors. AKT1 (ρ=+0.21, p=0.005) and FASN (ρ=+0.18, p=0.016) replicate; VPS35 and SCD do not.